MED 2181. Brain Sciences with Head and Neck Anatomy.
This is a core course for the ScM in Medical Sciences degree, part of the Gateways program at Alpert Medical School. This course builds on 4 required prerequisite courses offered in the Fall Semester. In this course, students learn the integration of neuroanatomy, gross anatomy and basic science principles, and the application of these principles to clinical neurologic dysfunction. In the anatomy portion of this course, students learn about the functional and developmental anatomy of the head and neck. This study is aided by review of prosections of the head, neck and brain, and builds on anatomical structure and function learned in the fall semester. The neurobiology section is designed to acquaint students with the major structures and functions of the nervous system, building on cell physiology and introductory materials from the fall semester, and the application to clinical dysfunction.

MED 2190. Microbiology and Infectious Disease.
Microbiology and Infectious Disease is an integrated course that introduces the basic biological principles, pathogenesis and host response, disease presentation, epidemiology, control and treatment of parasites, viruses, fungi and bacteria that cause human disease. Emphasis is placed on the most clinically significant and best characterized pathogens in each group. The Microbiology component of the course explores the characteristics of disease-causing microorganisms, mechanisms of transmission, immunity, and how specific microbial pathogens cause disease. Microbial disease states in multiple organ systems are addressed in the Infectious Disease component of the course with a focus on common infectious diseases and their clinical presentation.

MED 2200. Anatomy and Physiology.
This course will cover major organ systems and disease sites. Organ functions will be presented in addition to standard anatomy and cross-sectional imaging based on different modalities (x-ray Mammography, CT, MRI, PET, US). Organs at risk and dose tolerance to normal structures will be discussed. Image Registration and Fusion will also be covered, as will motion management.

MED 2210. Radiological Physics and Dosimetry.
This course will cover the fundamental physics behind radiation production and interaction, including a review of pertinent mathematics, classical mechanics, and nuclear physics. Topics to be covered within basic radiation physics: radioactive decay, radiation producing devices, characteristics of the different types of radiation (photons, charged and uncharged particles), mechanisms of their interactions with materials, and essentials of the determination of absorbed doses, by measurement and calculation, from ionizing radiation sources used in medical physics (clinical) situations.

MED 2220. Radiation Protection & Instrumentation.
This course examines principles of radiation protection with application to the hospital setting in radiation oncology, diagnostic imaging, and nuclear medicine. Designs of facilities and quality management programs are examined. Radiation safety practices are reviewed for involved hospital staff, patients, and the general public. This includes various radiation sources: electronically-generated photons and electrons, sources of sealed radioactivity, and unsealed sources of radioactivity. Additionally, the practice of radiation measurements as performed by the medical physicist is taught. This aspect includes associated dosimetry protocols, instrumentation, and clinical contexts. A practicum permits hands-on opportunities to assimilate the theoretical basis and rationale for radiation measurements.

MED 2230. Computational Medical Physics.
The aim of the Computational Medical Physics course is to familiarize students with mathematical, statistical and computational techniques in Medical Physics and how they integrate at a systems level. Students will learn about the emerging field of Computational Medical Physics through the application of mathematical modeling, computer simulations and quantitative and data-intensive analyses to medical data towards enhancing the accuracy, safety and efficiency of patient care and providing an understanding of cancer research. Basic programming skills are expected.

MED 2250. Radiation Therapy Physics.
This course will provide a comprehensive survey of basic radiotherapy physics, fundamental radiation therapy, and contemporary radiation therapy. The basic principles of radiotherapy treatment modalities, radiation detection, dose calibration methods, and image-based treatment planning will be reviewed. Topics to be covered include external beam radiation therapy (photons, protons, and electrons), brachytherapy, and special procedures. Image guidance methods will be discussed as well as patient and machine quality assurance.

MED 2260. Physics of Medical Imaging.
The course provides the necessary physics background that underpins day-to-day medical imaging physics activities. It is aimed primarily at new entrants to the profession, but should be of benefit to postgraduate students, postdoctoral research workers, physicist-managers, representatives of allied commercial organizations and anyone wishing to deepen or re-establish their understanding of the physics of medical imaging. Overviews of specialized or research related topics, such as positron emission tomography and magnetic resonance spectroscopy are given.

MED 2270A. Research and Clinical Practicum for Medical Physics.
Customized for each project Note : 2.5 Credits

MED 2280. Nuclear Medicine Physics.
Nuclear Medical Physics - PET

MED 2290. Advanced Radiation Therapy.
Advanced Therapy is meant to serve as a guided self-study of advanced / applied topics in radiation therapy with emphasis on current clinical usage. Optional topics include, but are not limited to, dose calculation algorithms, optimization techniques, deformable registration techniques, modeling within treatment planning systems, and treatment planning.

MED 2300. MR Imaging Technology, Ultrasound, and Interventional.
This course will provide an introduction to magnetic resonance imaging scanner hardware, image acquisition methods used in the clinical setting for various contrast weightings, imaging of physiologic function, and image reconstruction methods. Causes and corrective measures for image artifacts will be discussed. Image-guided interventions for therapeutic purposes are becoming increasingly common as minimally-invasive treatments increase in popularity. The course will discuss some common methods used in interventional techniques with attention to the hardware and real-time image acquisition methods used for such therapies. An introduction to ultrasound imaging will be given which will include the physical principles of image formation, application of real-time techniques, Doppler methods for assessing blood flow, and ultrasound use in interventional procedures.

MED 2310. Radiation Biology.
This program provides a comprehensive overview of radiation biology with a particular emphasis on aspects of direct relevance to the practice of radiation oncology. It addresses the molecular and cellular responses to radiation-induced damage that influence cell death in both tumors and normal tissues. Quantification of radiation effects and the underlying biological basis for fractionation of radiotherapy and dose-response relationships in the clinic are covered in depth. The biological basis for current approaches to improve radiotherapy will be described including novel fractionation schemes, retreatment issues, targeting hypoxia, and biological modifiers.

MED 2980. Independent Study in Population Medicine.
For students enrolled in the Primary Care-Population Medicine program at Alpert Medical School, this course is structured to allow students to conduct research focused on population health with a mentor at Brown University.